

DATA STRUCTURE

LAB PROGRAM

13. Write a C program to implement Queue operations such as ENQUEUE, DEQUEUE and Display

```
#include <stdio.h>
```

```
#define MAX 5
```

```
int queue[MAX];
```

```
int front = -1, rear = -1;
```

```
void enqueue(int item) {
```

```
    if (rear == MAX - 1) {
```

```
        printf("Queue Overflow\n");
```

```
    } else {
```

```
        if (front == -1)
```

```
            front = 0;
```

```
        queue[++rear] = item;
```

```
        printf("%d inserted into queue\n", item);
```

```
    }
```

```
}
```

```
void dequeue() {
```

```
    if (front == -1 || front > rear) {
```

```
        printf("Queue Underflow\n");
```

```
    } else {
```

```
        printf("Deleted element: %d\n", queue[front++]);
    }
}
```

```
void display() {
    int i;
    if (front == -1 || front > rear) {
        printf("Queue is empty\n");
    } else {
        printf("Queue Elements: ");
        for (i = front; i <= rear; i++) {
            printf("%d ", queue[i]);
        }
        printf("\n");
    }
}
```

```
int main() {
    int choice, item;

    do {
        printf("\n--- Queue Operations ---\n");
        printf("1. ENQUEUE\n");
        printf("2. DEQUEUE\n");
        printf("3. DISPLAY\n");
        printf("4. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```
switch (choice) {  
    case 1:  
        printf("Enter element: ");  
        scanf("%d", &item);  
        enqueue(item);  
        break;  
  
    case 2:  
        dequeue();  
        break;  
  
    case 3:  
        display();  
        break;  
  
    case 4:  
        printf("Exiting...\n");  
        break;  
  
    default:  
        printf("Invalid Choice\n");  
}  
} while (choice != 4);  
  
return 0;  
}
```

Output

```
--- Queue Operations ---
```

```
1. ENQUEUE
```

```
2. DEQUEUE
```

```
3. DISPLAY
```

```
4. EXIT
```

```
Enter your choice: 11
```

```
Invalid Choice
```

```
--- Queue Operations ---
```

```
1. ENQUEUE
```

```
2. DEQUEUE
```

```
3. DISPLAY
```

```
4. EXIT
```

```
Enter your choice: 1
```

```
Enter element: 3
```

```
3 inserted into queue
```

```
--- Queue Operations ---
```

```
1. ENQUEUE
```

```
2. DEQUEUE
```

```
3. DISPLAY
```

```
4. EXIT
```

```
Enter your choice:
```